

Cross-Chain Content Rights Management Service: A Framework for Decentralized Subscription, Pay-Per-View, and Purchase Models

Abstract

This research proposal delineates a comprehensive investigation into the development of a cross-chain content rights management service. Unlike traditional Digital Rights Management (DRM) systems, which predominantly emphasize content security and access control, this study focuses on the monetization and dissemination of content rights through decentralized models such as subscription, pay-per-view, and purchase options. The proposed framework aims to establish an interoperable, scalable, and transparent ecosystem that accommodates content creators, consumers, and distributors across various blockchain networks.

1 Introduction and Motivation

The digital content industry has witnessed unprecedented growth, with global digital media revenues projected to surpass \$370 billion by 2025. However, the prevailing centralized framework for content distribution faces considerable challenges, including elevated intermediary fees, restricted control for creators, geographical limitations, a lack of interoperability, and unclear revenue-sharing mechanisms. Content creators frequently surrender significant portions of their earnings to centralized platforms, while consumers face fragmented engagement across various platform services.

Traditional content rights management systems primarily concentrate on preventing unauthorized access through Digital Rights Management (DRM) technologies. However, a significant gap persists in the effective management of the commercial dimensions of content rights—specifically subscription management, pay-per-view transactions, and purchase verification—across various platforms and blockchain networks. The advent of blockchain technology, particularly interoperable frameworks, offers a promising opportunity to transform content rights management by creating decentralized, transparent, and efficient monetization mechanisms.

The motivation for this research arises from several fundamental considerations. Primarily, the shift towards the Web3 paradigm emphasizes the significance of user ownership and creator empowerment, thereby necessitating the development of innovative models for content monetization that are aligned with decentralized principles. Additionally, the fragmented landscape of existing blockchain ecosystems necessitates the implementation of cross-chain solutions to facilitate seamless content access and efficient payment processing. Furthermore, the growing adoption of subscription and micro-payment models in digital content consumption underscores the need for advanced, programmable rights management systems that can function autonomously across multiple platforms.

2 Research Question

What are the most effective strategies for designing and implementing a cross-chain content rights management service that aims to facilitate decentralized subscription, pay-per-view, and purchase models while ensuring scalability, interoperability, and economic efficiency across multiple blockchain networks?

Sub-questions:

- How might cross-chain protocols be employed to facilitate seamless transactions of content rights across multiple blockchain networks?
- What are the most effective design patterns for balancing decentralization with user experience in content rights management?

3 Background and Related Work

3.1 Blockchain-Based Content Management

Recent research has examined various methodologies for blockchain-enabled content management and copyright protection. Xie et al. (2022) introduced a multi-blockchain framework designed to safeguard digital copyright, utilizing cross-chain technology. This framework encompasses three concurrent chains, each designated for digital work management, copyright administration, and dispute resolution. Their findings demonstrate improved security measures and reduced block creation times compared to single-chain solutions, thereby highlighting the potential of cross-chain architectures in content management applications.

Similarly, research on blockchain cross-chain technology for digital resource copyright protection has demonstrated promising outcomes in enhancing system maintainability

and scalability, while more effectively aligning with real-world business requirements in comparison to single-chain or consortium blockchain methodologies. These investigations provide foundational insights into the technical feasibility of cross-chain content management systems.

3.2 Cross-Chain Interoperability

The development of cross-consensus messaging standards represents significant progress in blockchain interoperability. These standards serve as messaging formats rather than merely as cross-chain protocols, intending to facilitate communication among various consensus mechanisms, including blockchains, smart contracts, and bridged networks. Research has demonstrated that the architecture of these standards facilitates programmable cross-chain interactions, including functionalities such as asset transfers, remote execution, and multi-hop process communication.

Recent implementations of cross-consensus messaging have demonstrated significant adoption, with high volumes of messages processed across interconnected networks. The flexibility allows for the use of various transport methods, thereby presenting different trade-offs between efficiency and resource utilization.

3.3 Smart Contract Platforms and Applications

The evolution of smart contract platforms has been thoroughly analyzed, with particular emphasis on their applications across various industries. Research has outlined six primary areas of smart contract development: technical foundations, integration with the Internet of Things, security and verification mechanisms, potential for industry disruption, associated challenges and opportunities, and legal considerations implications.

Recent analyses of smart contract research indicate a growing interest in domain-specific applications, with content distribution and digital rights emerging as prominent areas of examination. The integration of smart contracts with content distribution networks has demonstrated potential for establishing more transparent, automated, and efficient mechanisms for content monetization models.

3.4 Blockchain Architectures and Smart Contracts

Blockchain architectures that provide shared security to interconnected chains offer unique advantages for cross-chain applications. The accommodation of diverse virtual machine environments enhances adaptability for various use cases.

Domain-specific languages for smart contracts have garnered significant popularity within interoperable blockchain ecosystems. Research indicates that these contracts offer numerous benefits, such as enhanced memory safety, efficient execution, and seamless

integration with runtime modules. The capability to prototype contracts before migration to dedicated chains provides a well-organized development pathway conducive to scalable applications.

3.5 Decentralized Content Monetization

Recent scholarly investigations into blockchain-based monetization of digital content have examined various models of tokenization. These studies have focused on how content creators can tokenize diverse media formats—such as text, audio, video, and interactive media—thereby facilitating innovative monetization mechanisms, including micro-transactions, subscription tokens, and fractional ownership models.

The concept of Non-Fungible Tokens (NFTs) has been extensively analyzed as a means of establishing digital ownership and enabling content monetization. However, research indicates that current NFT marketplaces predominantly focus on collectibles rather than practical applications in content distribution and rights management, thereby revealing a gap that this research aims to address.

4 Proposed Methodology/Approach

This research intends to utilize Polkadot’s ecosystem as the foundational blockchain infrastructure, selected for its robust cross-chain interoperability and shared security model, which surpasses single-chain alternatives such as Ethereum in scalability and cost-efficiency (Wood, 2016). The development process will employ the Rust programming language for optimal performance, ink! for the secure implementation of smart contracts, and Cross-Consensus Message Passing (XCM) protocols to facilitate seamless cross-chain interactions. The modular framework consists of three primary components:

1. Subscription contracts for recurring access
2. Pay-per-view contracts for one-time transactions
3. Purchase verification contracts for ownership transfers

all integrated through chain extensions with Substrate runtime modules to ensure efficient state management.

The methodology adheres to an iterative prototyping process encompassing three phases:

1. Developing and testing ink! contracts on a local testnet parachain utilizing Dockerized nodes

2. Integrating XCM to facilitate cross-chain rights transfers between parachains and external blockchains such as Ethereum via bridges
3. Refining Rust components to attain low latency and high throughput through techniques such as batch processing and caching.

Scalability shall be augmented by migrating prototypes to dedicated parachains, capitalizing on Polkadot’s slot-based architecture to accommodate increased transaction volumes. Potential challenges, including XCM latency or vulnerabilities in smart contracts, shall be addressed through comprehensive static analysis (**cargo-contract**) and fuzzing (**cargo-fuzz**) to ensure secure and efficient operations. This strategy establishes a scalable, interoperable framework that also considers decentralized monetization requirements.

5 Evaluation

The proposed cross-chain content rights management service shall be evaluated employing a combination of quantitative metrics to assess its efficacy, scalability, interoperability, economic efficiency, security, and the equilibrium between decentralization and centralization. The evaluation will be conducted within simulated environments—such as local development configurations utilizing Dockerized blockchain nodes—as well as in genuine testnet environments, including platforms like Polkadot’s Westend or Kusama testnets. Comparisons will be made against centralized alternatives, such as conventional Digital Rights Management (DRM) platforms exemplified by Adobe Content Server, or centralized subscription services akin to Netflix’s backend systems.

To ensure a comprehensive assessment, the evaluation will adhere to a phased approach:

1. Unit and integration testing during development
2. Simulated stress testing in controlled environments
3. Deployment to testnets for real-world-like conditions
4. Iterative analysis of results.

Tools such as performance profiling software (e.g., Rust’s **cargo-flamegraph** or **perf**), automated testing frameworks (e.g., Substrate’s testing utilities), blockchain explorers (e.g., Subscan for transaction monitoring), and metrics collectors (e.g., Prometheus and Grafana) will be employed. The results will be validated against benchmarks from related works, such as those outlined in Xie et al. (2022) for cross-chain performance, as well as against baseline metrics for centralized systems. Iterative improvements will

be implemented to ensure that the framework aligns with decentralized principles while maintaining optimal performance and competitiveness.

Key aspects to evaluate include:

- **Interoperability and Transaction Seamlessness:** Evaluate the success rate (percentage of successful transfers) and latency (average duration from initiation to confirmation) of cross-chain content rights transfers, such as subscription activations or pay-per-view accesses across blockchain networks. Stress testing will involve automated scripts (e.g., employing Rust’s `subxt` library) to simulate transaction volumes ranging from 10 to over 1,000 per minute, with metrics collected via Prometheus (e.g., `xcm_message_processed_counter`) and visualized in Grafana. For comparative purposes, centralized systems such as Adobe Content Server will be simulated utilizing AWS API Gateway and Lambda, aiming for approximately 200 milliseconds latency for license requests (based on REST API standards). For instance, if the proposed system attains a 99% success rate and 2-second latency, whereas AWS achieves 95% success and 200 milliseconds latency, considerations regarding the trade-offs between speed and decentralization will be analyzed quantified.
- **Scalability:** Assess system performance under increasing loads, such as thousands of concurrent subscriptions or purchases, using load generators (e.g., Rust-based scripts with `tokio` or `k6`). Metrics include throughput (transactions per second, TPS), resource utilization (e.g., CPU/memory on validator nodes, monitored via Prometheus), and block creation times (via Subscan). Tests will scale node counts from 5 to 50 on testnets, with performance curves plotted to identify bottlenecks. Centralized benchmarks will employ AWS EC2 instances (e.g., `t3.large`) simulating Netflix-like subscription APIs, aiming for approximately 5,000 TPS (as referenced by Netflix technical blog). For example, if the system attains 500 TPS with 60% CPU utilization, compared to AWS’s 5,000 TPS with 80% CPU, scalability trade-offs will be analyzed.
- **Economic Efficiency:** Analyze cost-effectiveness by calculating transaction fees (in native tokens, converted to USD), revenue-sharing transparency (verified on-chain via Subscan), and cost savings for creators/consumers compared to centralized platforms (e.g., 30% fees in Adobe Content Server versus less than 5% in the proposed system). Simulations will model monetization scenarios (subscription, pay-per-view, purchase) employing Monte Carlo methods in Rust or Python, with variations in user counts and gas prices. Centralized costs will be estimated using the AWS Pricing Calculator (e.g., \$0.05 per API call for a Netflix-like backend). For example, if the system achieves \$0.02 per transaction compared to AWS’s \$0.05, it demonstrates a 60% reduction in savings.

- **Security and Reliability:** Conduct audits and penetration testing on smart contracts and cross-chain protocols utilizing tools such as `cargo-contract` for static analysis and `cargo-fuzz` for fuzz testing. Metrics encompass resistance to attacks (including reentrancy, unauthorized access, assessed via exploit success rate) and uptime during simulated failures (such as node crashes simulated with Chaos Mesh in a Kubernetes-based testnet). Mean time to recovery (MTTR) will be documented, with a target of less than one minute. Centralized systems will be emulated (for instance, AWS Lambda with AES encryption employed for digital rights management) and examined for vulnerabilities (such as API key disclosures via Burp Suite), with uptime performance compared against AWS Service Level Agreements (approximately 99.9%). For example, if the system attains 99.95% uptime in comparison to AWS’s 99.9%, the resilience of decentralization is demonstrated and highlighted.
- **Decentralization Balance:** Quantify decentralization through the analysis of node distribution, utilizing the Herfindahl-Hirschman Index (HHI) computed via Python scripts (e.g., `numpy` for stake calculations). Experimental evaluations will be conducted with variable node counts ranging from 5 to 50 on test networks, correlating HHI with performance metrics such as latency and Transactions Per Second (TPS) monitored via Prometheus. Centralized systems, such as Netflix’s AWS-based backend, exhibit an HHI of approximately 10,000, indicative of control by a single entity, thus establishing a reference point for comparative purposes. For example, If the system attains an HHI of 1,200 with a latency of 1.5 seconds, compared to AWS’s HHI of 10,000 and latency of 200 milliseconds, this reflects an optimal balance between decentralization and performance.

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